Quantum Approximate Optimization Algorithm

Application to Financial Stock Portfolio Optimization

FYS5419 Project 2

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Abstract

Financial markets exhibit pronounced non-stationarities driven by structural breaks, macroeconomic shocks, and evolving investor behavior. This project investigates the use of machine learning methods for identifying and modeling regime changes in financial time series. We consider supervised and unsupervised approaches to regime detection, including clustering-based methods and probabilistic state models, and evaluate their ability to capture shifts in volatility, return dynamics, and cross-asset dependencies. Model performance is assessed using historical market data under realistic out-of-sample settings. The results highlight both the potential and limitations of machine-learning-based regime modeling in quantitative finance, with implications for risk management, portfolio allocation, and adaptive trading strategies.

1 Introduction

This is an intro lorem ipsum with a reference Hastie et al. (2009) and such (Hastie et al. 2009).

Quantum computing combines ideas from linear algebra, probability theory, and computer science to enable new computational paradigms beyond classical algorithms. Introductory expositions such as Rieffel and Polak (1998) and Olivares (2020) explain how qubits, superposition, and entanglement form the foundation of quantum information processing, while more mathematically oriented treatments emphasize operator theory and Hilbert space methods (Scherer 2019). From a programming perspective, practical workflows and software abstractions are discussed in Hundt (2022) and Hidary (2019).

One important application area is quantum machine learning, where hybrid classical–quantum models and variational circuits are increasingly studied (Schuld and Petruccione 2018; Conti 2023; Pattanayak 2021). These methods often rely on variational optimization frameworks similar to those used in near-term quantum algorithms.

A prominent example is the Quantum Approximate Optimization Algorithm (QAOA), which provides a variational approach to solving combinatorial optimization problems on noisy intermediate-scale quantum (NISQ) devices. The theoretical structure and implementation considerations of QAOA are described in Zhou et al. (2018), while introductory and experimental perspectives can be found in (Giovagnoli 2020; Irfan 2023). Together, these works illustrate how hybrid optimization loops between classical and quantum processors form a central paradigm in modern quantum algorithm design.

More broadly, quantum computation and quantum information theory provide the conceptual framework linking these developments, including complexity-theoretic motivation and algorithmic primitives such as amplitude amplification and phase estimation (Michaelson and Isaac 2010).

The QAOA workflow can also be implemented using modern quantum SDKs such as Qiskit, which provide end-to-end tutorials covering circuit construction, parameter optimization, and execution on simulators or hardware (IBM Quantum 2024).

2 Methods

This is the methods section, lorem ipsum.

For NBIM, might be relevant with: Constrained Portfolio Optimization using Variational Quantum Eigensolvers (VQE) or QAOA.

The Method:Formulate the "Minimum Volatility" problem. Instead of maximizing $Return - Risk$, you simply minimize $w^T \Sigma w$ (where w is weights, Σ is the covariance matrix).The Twist: Add a constraint that makes it hard for classical computers, such as "Cardinality Constraints" (e.g., "Select exactly K assets out of N that give the lowest total volatility").The Quantum Part: Map this to a QUBO (Quadratic Unconstrained Binary Optimization) problem and solve it using VQE.

The Dataset Construction:Segment A: The Global Financial Crisis (Sept 2008 – Mar 2009) $\rightarrow$ 150 days of pure panic.Segment B: The Euro Debt Crisis (Selected months in 2011) $\rightarrow$ 50 days of high volatility.Segment C: The Covid Crash (Feb 2020 – April 2020) $\rightarrow$ 50 days of extreme speed.Segment D: The Inflation Shock (2022) $\rightarrow$ 100 days of slow grind.Total Data: $\tilde{3}$00–400 data points (Daily Returns).Why this works:Perfect Size: 400 points is perfect for a Quantum Circuit Born Machine (QCBM). It's enough to define a probability distribution but small enough to train quickly on a simulator.Signal-to-Noise: You have removed the 90% of history where nothing happens. Your model will learn a "Fat Tailed" distribution (high probability of extreme events), which is the Holy Grail of risk management.2. The Granularity Issue (Daily

vs. Sub-daily)Stick to Daily Data.Reason 1: You cannot get free, reliable 1-minute data from 2008. It doesn't exist in public APIs.Reason 2: Sub-daily data is "noisy." A 1-minute crash is often just a glitch. A daily crash (-5% close-to-close) is a macroeconomic event.The "Pro" Move: If you want to show off, use "Log Returns" rather than raw prices.

$R_t = \ln(\frac{P_t}{P_{t-1}})$

This normalizes the price difference (so the S&P 500 at 900 points in 2008 is mathematically comparable to S&P 500 at 5000 points in 2026).

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NBIM's report admits that most of history is irrelevant for stress testing.The "Water": 95% of market history is boring. Prices go up 0.1% or down 0.1%. This data contains almost zero information about risk.The "Poison": The crashes (2008, 2020) are where the physics of the market change.Your Strategy: By stitching together only the crashes (approx. 400–500 days), you are creating a dataset that is 100% signal and 0% noise. You are training the quantum computer purely on the "physics of disaster."In your report: You frame this as "Regime-Specific Training." You aren't modeling "The Stock Market"; you are modeling "The Crisis Regime." This is a sophisticated econometric technique, not a shortcut.2. How to make the project "Bigger" (The Multi-Asset Twist)If you are worried that just predicting "The S&P 500" is too simple, you can instantly make the project "cracked" by adding dimensions.NBIM's report (Source 266, Chart 9) shows losses across Equities, Fixed Income, and Real Assets.Instead of feeding your quantum circuit just one number (Stock Return), feed it a Vector of 3 numbers for each day:Equity Return (S&P 500)Bond Return (US 10Y Treasury)Oil Price Return (Brent Crude - highly relevant for Norway)Now the math gets hard:Input: 400 days $\times$ 3 assets = 1,200 data points.The Challenge: The Quantum Circuit now has to learn not just the probability of a crash, but the Correlation Structure during a crash.Normal times: Stocks up, Bonds down (Negative correlation).Crash times: Stocks down, Oil down, Bonds ?? (Correlations often break or flip).The Win: If your QCBM can successfully generate synthetic data where Stocks and Oil crash together (which hurts NBIM the most), you have effectively modeled the "Fragmented World" scenario mentioned in their report.+13. The "Big Data" is in the Output, not the InputThe value of a Generative Model (like ChatGPT or your QCBM) isn't the size of the training set; it's the infinite amount of unique data it can produce.Input: 500 days of historical crashes.Output: You will generate 10,000+ new crash days for your simulation.The "Flex": In your 10-page report, you don't show a table of the 500 input days. You show a 3D Density Plot of the 10,000 generated days, comparing it to the historical distribution to prove your quantum model "learned" the shape of the risk.4. Summary: The "Cracked" Semester Project PlanHere is the final, confident outline for the "Black Swan" project:Title: Generative Quantum Learning for Multi-Asset Stress Testing of the GPFGData Curation (The "Museum of Disasters"):Source daily returns for Equities (SPY), Bonds (TLT), and Energy (XLE).Filter for high-volatility regimes: 2008 (Financial Crisis), 2020 (Covid), 2022 (Inflation/War).Total Data: 500 days of "Crisis Vectors."The Quantum Model (QCBM):Use a Variational Quantum Circuit (VQE/QCBM).Ansatz: Use RealAmplitudes or TwoLocal circuit from Qiskit.Training: Minimize the distance (KL Divergence) between the quantum distribution and your "Crisis Data."The Stress Test (The "Application"):Generate 1,000 synthetic "Future 2026" timelines.Calculate the Expected Shortfall (ES) for a portfolio holding these 3 assets.Compare it to NBIM's reported ES to see if your quantum model predicts larger potential losses (i.e., is it more conservative/safe?).+1Conclusion:"NBIM's current stress tests rely on historical replay. This project demonstrates that Quantum Generative AI can synthesize novel correlation breakdowns, providing a more rigorous test for 'Black Swan' events."

2.1 Quadratic Unconstrained Binary Optimization (QUBO)

$$\min_{x \in \{0,1\}^n} \left(q x^T \Sigma x - \mu^T x + \lambda (1^T x - B)^2 \right)$$

where q is risk tolerance, B is budget/number of assets to pick, and λ is a massive penalty factor to ensure the algorithm respects the budget.

3 Results

These are the results

4 Conclusion

References

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5 Appendix

Table 1. Influential Figures and Their Seminal Contributions

Figure	**Field**	**Most Famous Work / Contribution**
Max Planck	Quantum Mechanics	Discovery of energy quanta ($E = h\nu$)
Niels Bohr	Quantum Mechanics	The Bohr Model of the atom and Complementarity
Werner Heisenberg	Quantum Mechanics	The Uncertainty Principle and Matrix Mechanics
Erwin Schrödinger	Quantum Mechanics	The Schrödinger Equation and Wave Mechanics
Richard Feynman	QM / Computing	Path Integral Formulation and Quantum Simulators
John Bell	Quantum Foundations	Bell's Theorem (Violating Local Realism)
Seth Lloyd	Quantum ML	Quantum Principal Component Analysis (QPCA)
Peter Shor	Quantum Algorithms	Shor's Factoring Algorithm (Exponential Speedup)
Maria Schuld	Quantum ML	Quantum Kernels and Variational Quantum Circuits
Aram Harrow	Quantum ML	The HHL Algorithm for Linear Systems of Equations